

Analytical Finance

Macro-Economic Blame Event Study



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Introduction

Project Overview

Our goal in this class is to form a hypothesis as to why firms may experience abnormal returns based on their earning calls transcript (tackling this traditionally structured data problem from an unstructured angle) and create an event study to investigate our hypothesis.

Data Overview

As a class, we pulled earning calls data from Capital IQ of all US-based firms from 2010-2024. After cleaning, this resulted in ~189k earning calls transcripts.

6,294

Number of companies

5.7%

*Irregularity Rate
(company years with <4
transcripts / total
company years)*

5.2%

*Average % of sentences
with macro related words*

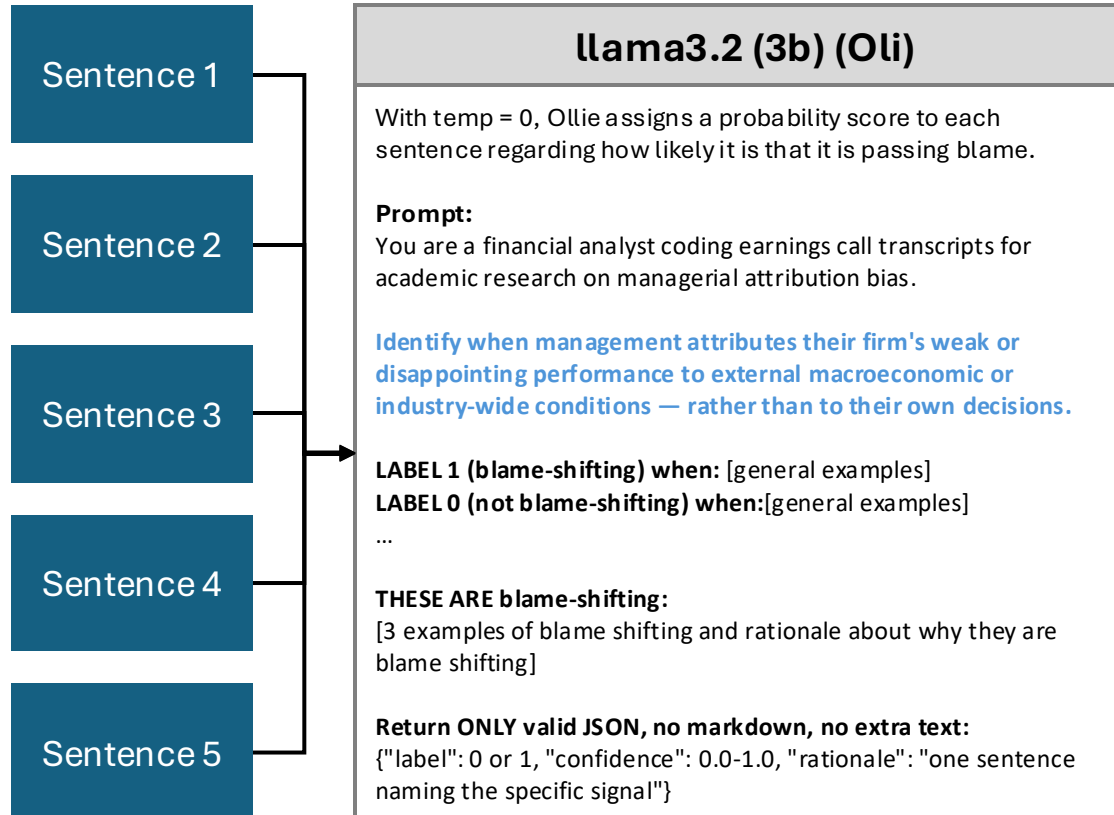
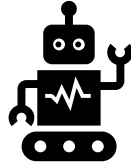
Hypothesis

Key Argument

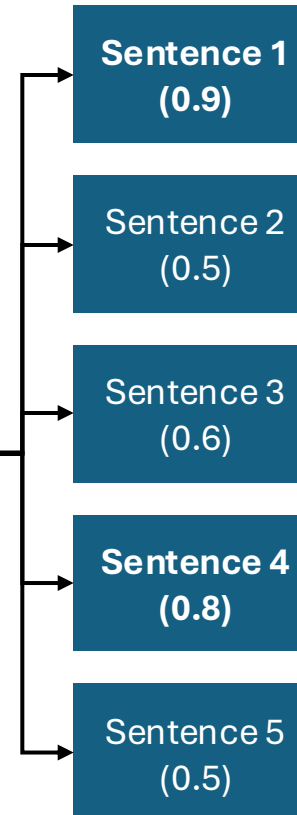
- **Premise 1:** Investors use earnings calls to **form views about a company**. The language management uses, especially how they explain performance, **provides key signals for investors**.
- **Premise 2:** When management **attributes performance to macroeconomic factors**, investors interpret this as a **signal about managerial quality and firm-specific conditions**.
- **Inference 1:** If a firm blames the macroeconomy more than other firms in the same industry, investors may view this as **failing to take accountability for poor performance**.
- **Inference 2:** This perception may lead investors to **believe underperforming areas will be ignored** and update their expectations about its future performance.
- **Inference 3:** As investors update their beliefs when this information is revealed during the earnings call, **stock prices adjust accordingly**.
- **Conclusion (Event Study Hypothesis):** Firms that **increase macro-economic blame** more than their industry peers in a given quarter tend to experience **greater negative cumulative abnormal returns (CAR)** following earnings calls.

Fine-tuning FinBERT

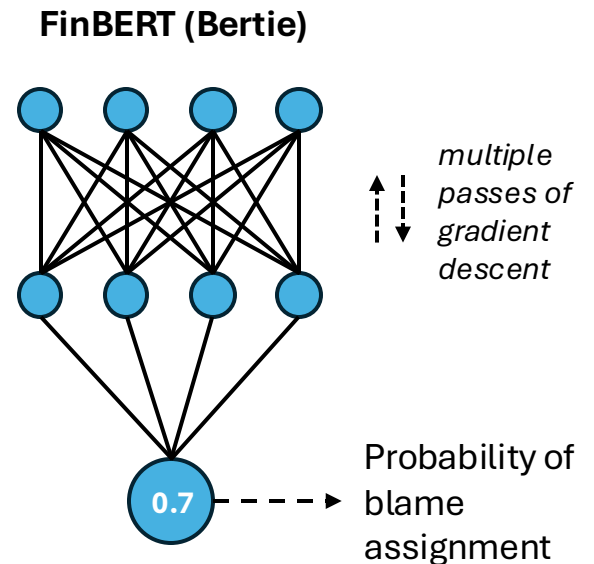
First, we extracted 22k sentences containing words from our macro-dictionary.



Confidence cutoff:
0.7

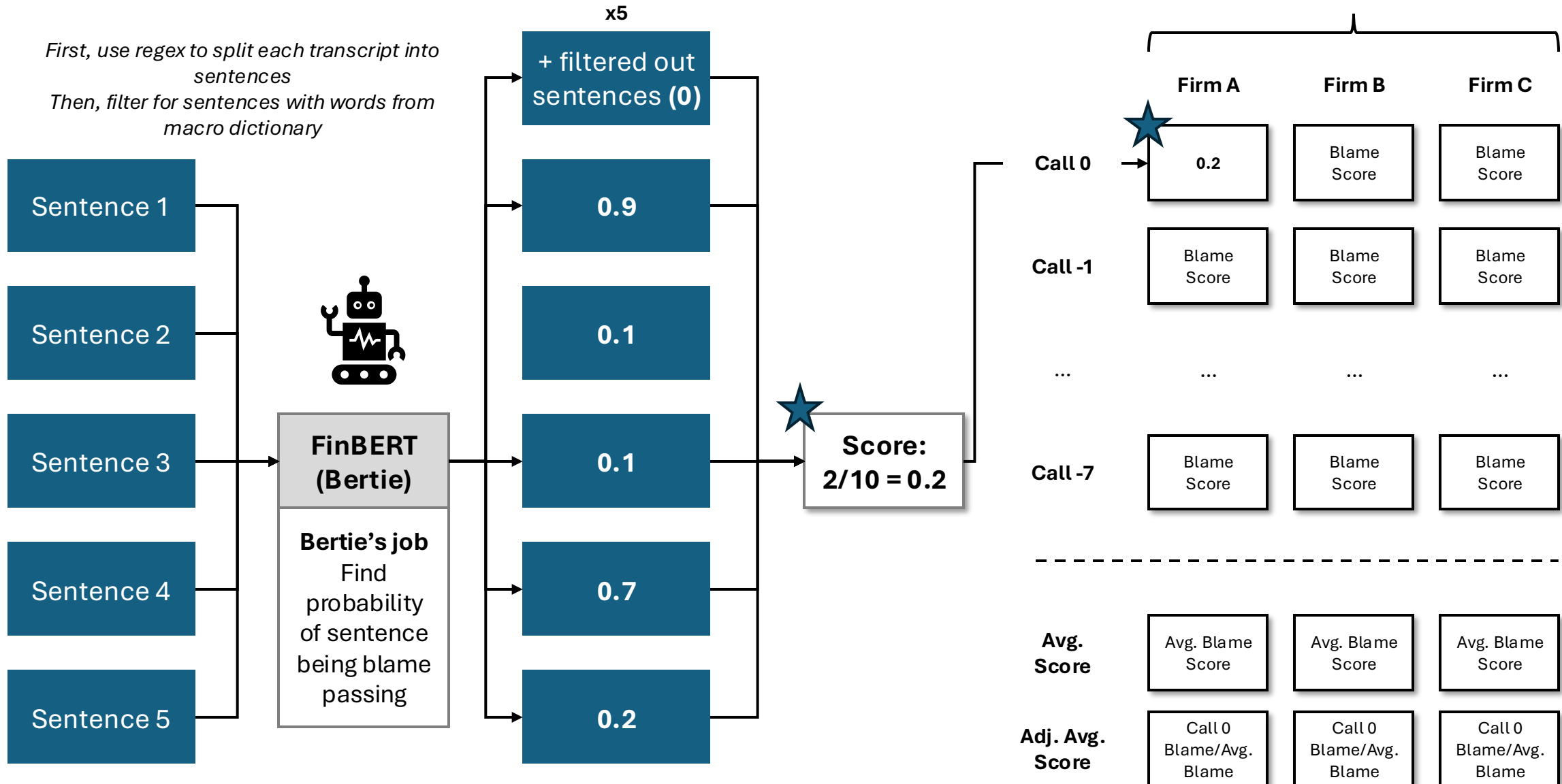


Then, using the sentences above the confidence cutoff, we **fine-tune FinBERT sentiment** via gradient descent to predict the probability of blame assignment rather than probability of positive sentiment.



Oli: 3b parameters
Bertie: 110m parameters
(27x smaller)

Blame Score Calculation



Blame Score Calculation

	Firm A	Firm B	Firm C
Call 0	 0.2	Blame Score	Blame Score
Call -1	Blame Score	Blame Score	Blame Score
...	...	...	...
Call -7	Blame Score	Blame Score	Blame Score

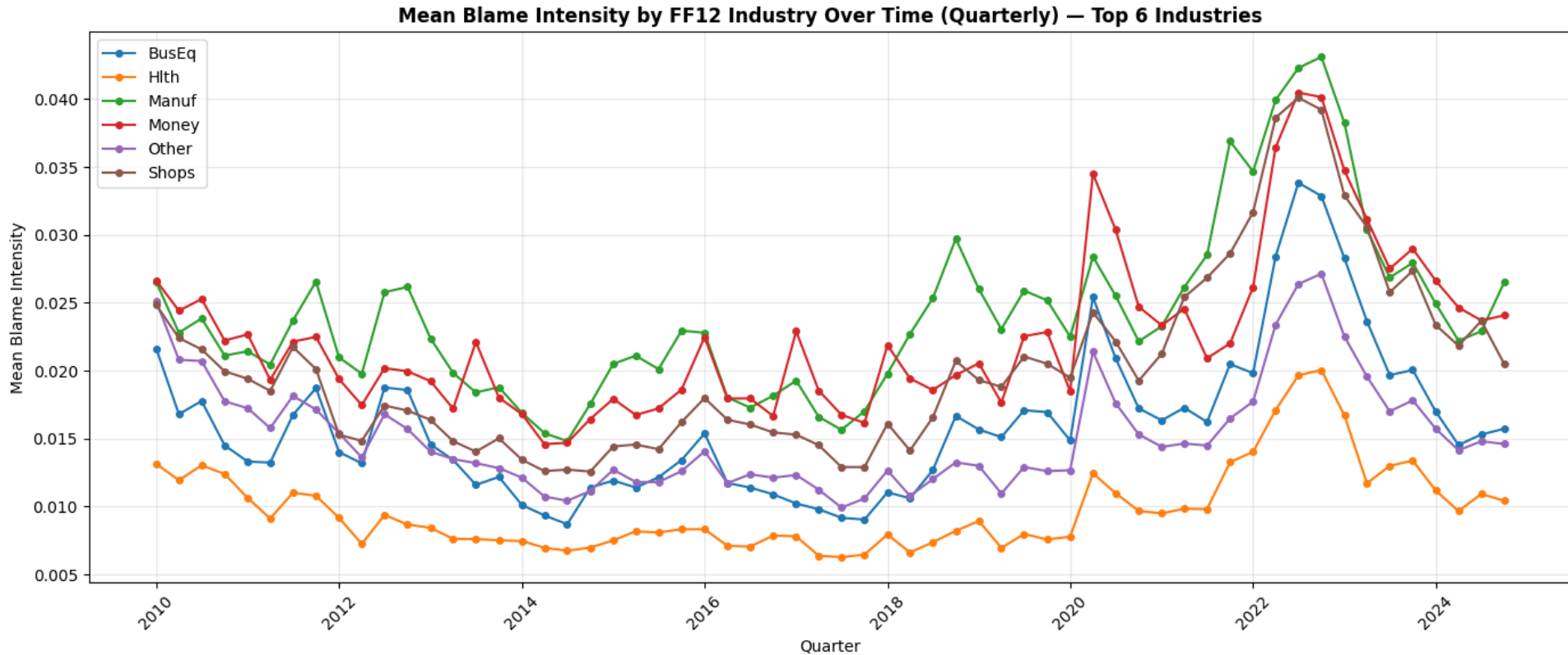
Avg. Score	Avg. Blame Score	Avg. Blame Score	Avg. Blame Score
Adj. Avg. Score	 Call 0 Blame/Avg. Blame	 Call 0 Blame/Avg. Blame	 Call 0 Blame/Avg. Blame

FINAL SCORE FOR FIRM A CALL 0

$$\textcircled{1} - \text{Avg} (\textcircled{2} + \textcircled{3})$$

1. Firms with less than 3 peers are excluded.
2. Final score subtracts the average rather than dividing as adj. avg. score is already normalized.

EDA – Mean Blame Intensity

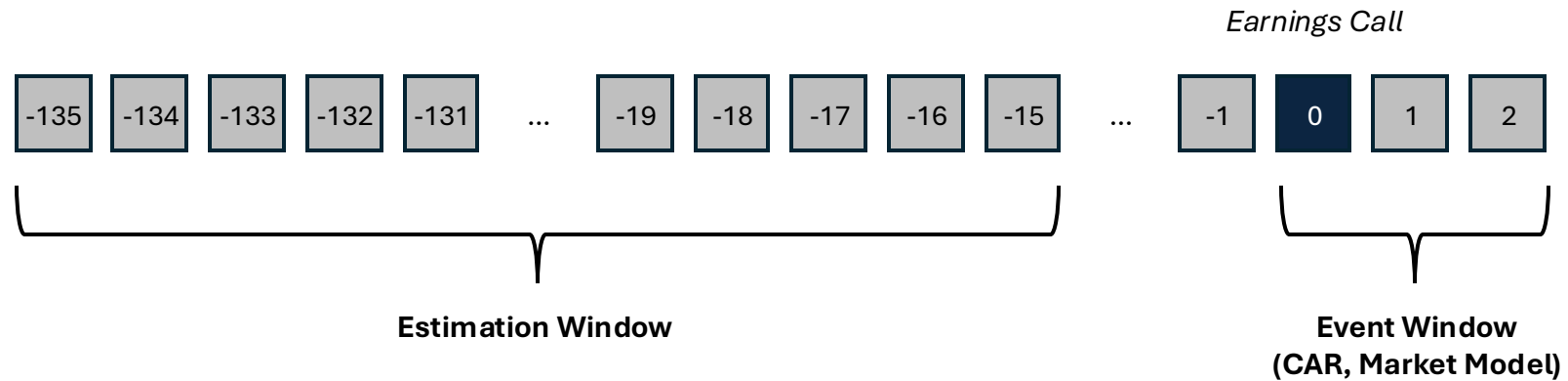


Key Takeaway

- All the industries follow a similar trend and peak around 2020-2023 (when the macroenvironment was the most volatile).
- This is evidence for our blame score accurately reflecting when blame is being shifted.

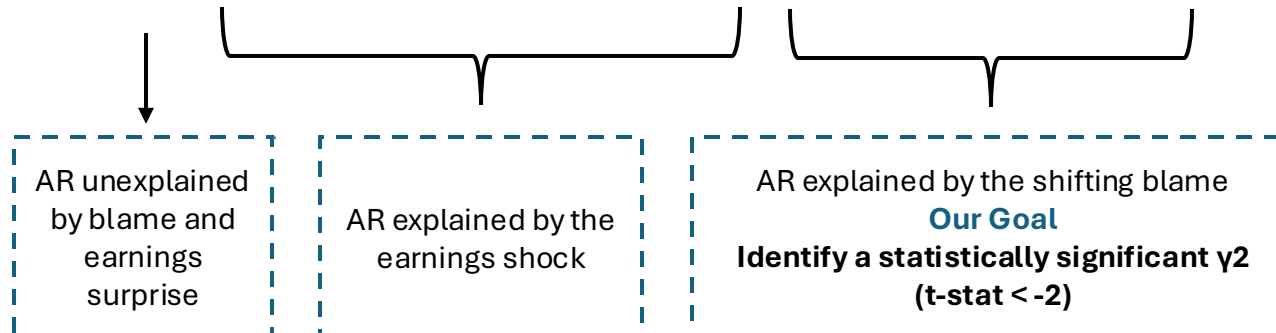
Event Study Design

Estimation and Event Windows

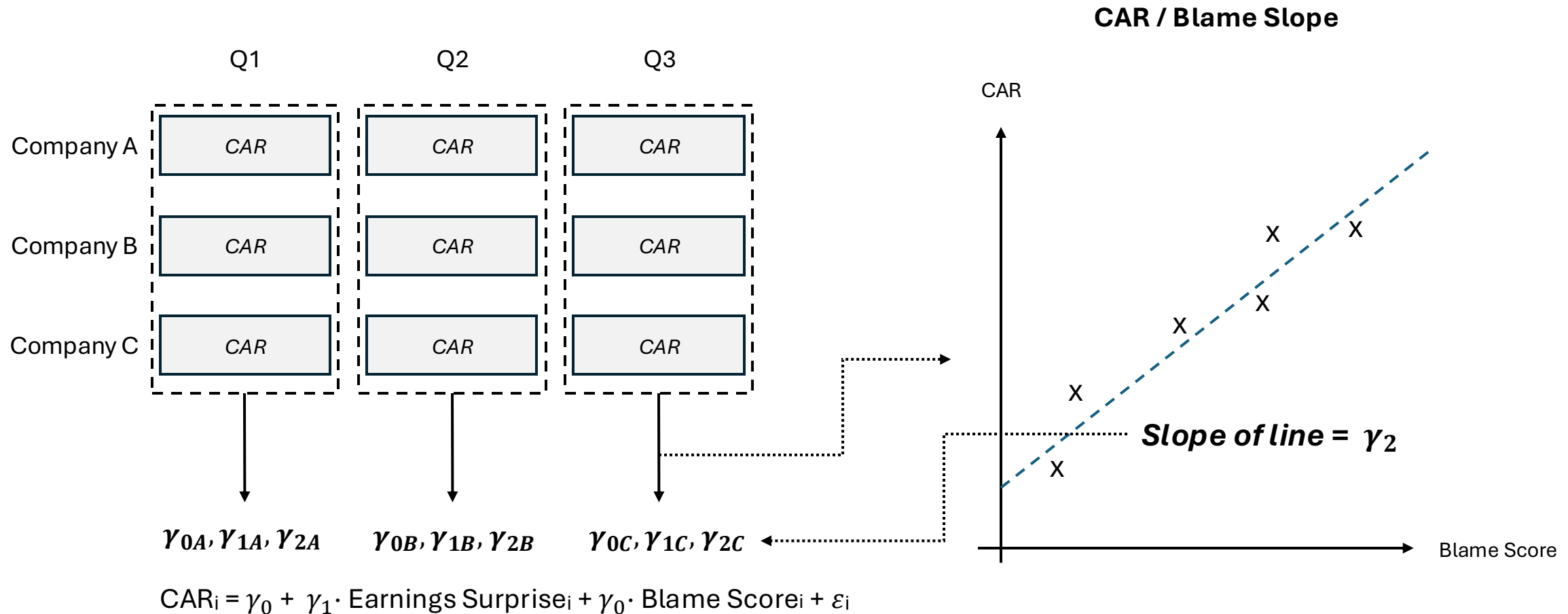


Regression Equation

$$CAR_i = \gamma_0 + \gamma_1 \cdot \text{Earnings Surprise}_i + \gamma_2 \cdot \text{Blame Score}_i + \varepsilon_i$$



Model Explanation – Fama-Macbeth



Run a t-test on the list of gammas returned

Results – Fama Macbeth Regression

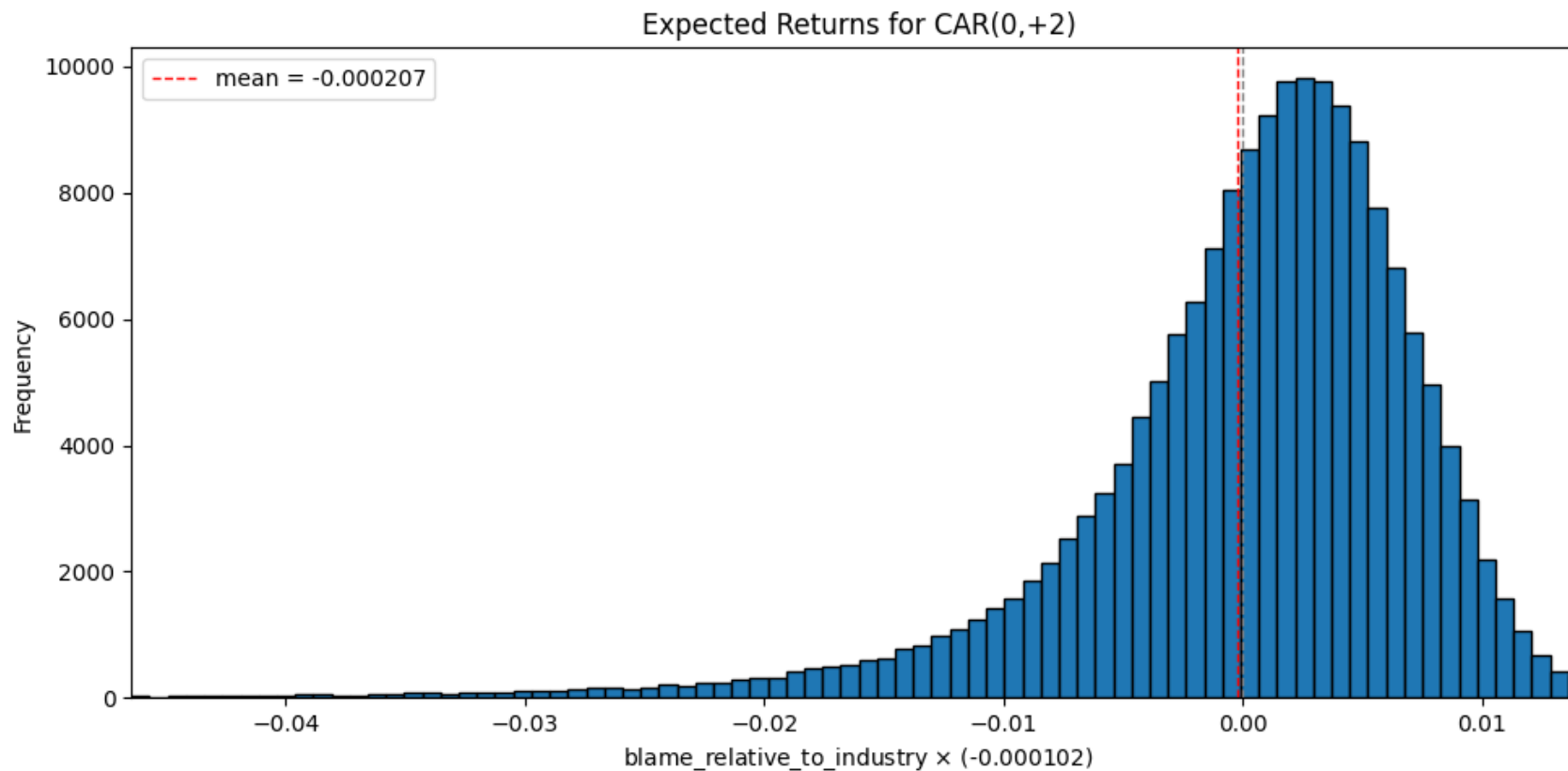
N = 140,600		Coefficient	t-stat	p-value
Y ₀	Intercept	-0.003172	-3.575	0.0003
Y ₁	Earnings Surprise	0.002578	4.567	0.0001
Y ₂	Blame Score	-0.000102	-8.512	0.0000



Key Takeaway

- The t-stat for earnings surprise is significantly positive, as is expected based on literature.
- The t-stat for blame score is significantly negative, which supports our hypothesis.

Returns Estimated from Blame



Key Takeaway

- Mean return of -0.02%, standard deviation of 1.2%, IQR of [-0.3%, 0.5%]

Conclusion

1

Hypothesis Supported

- Significant negative relationship with blame shifting (t-stat of -8.5)

2

Limited Economic Power

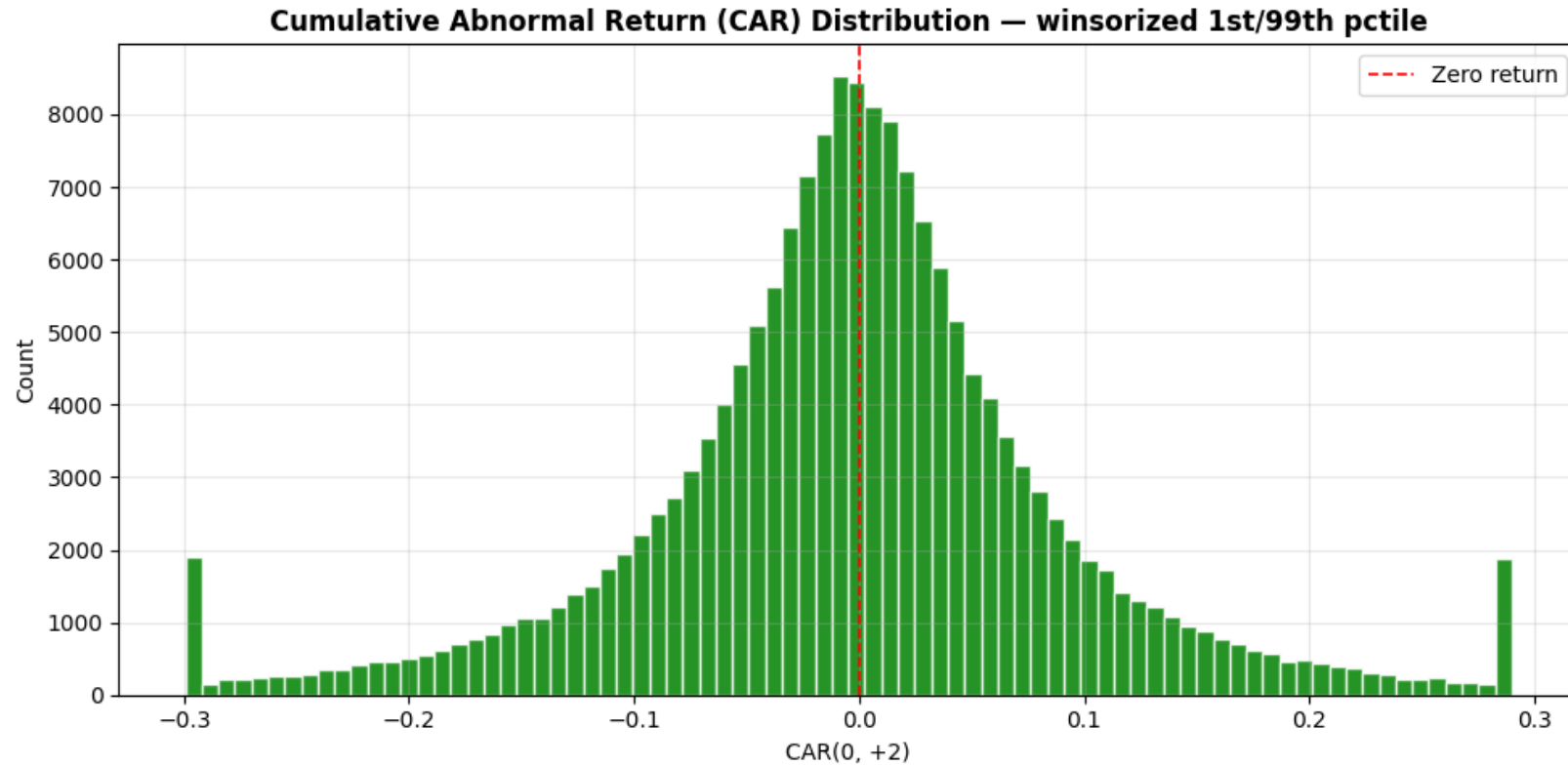
- (Blame Score $\cdot \gamma_2$) typically in $[-0.3\%, 0.5\%]$ range
- Given noise in short term CAR, it may be difficult to trade on

3

Extensions

- Use as one signal in a trading strategy
- Test for exaggerated effects under more specific conditions
- Investigate longer event windows

Appendix – CAR Distribution



Key Takeaway

- Predicting future CAR can result in a double-digit percentage returns
- Mean = -0.0024, median = -0.0017, std = 0.1071

Appendix - Checking FinBERTs Results

Shifting Blame:

Gross margin of 38.2% was down 180 basis points year-over-year. **About half of that was input cost inflation — we're still lapping elevated resin and freight costs — and the remainder reflects deleverage on the lower volume.** We're doing what we can on price, but in this demand environment, the elasticity just isn't there the way it was eighteen months ago.

Probs : {'negative': 0.0643, 'positive': 0.9357}

Not Shifting Blame:

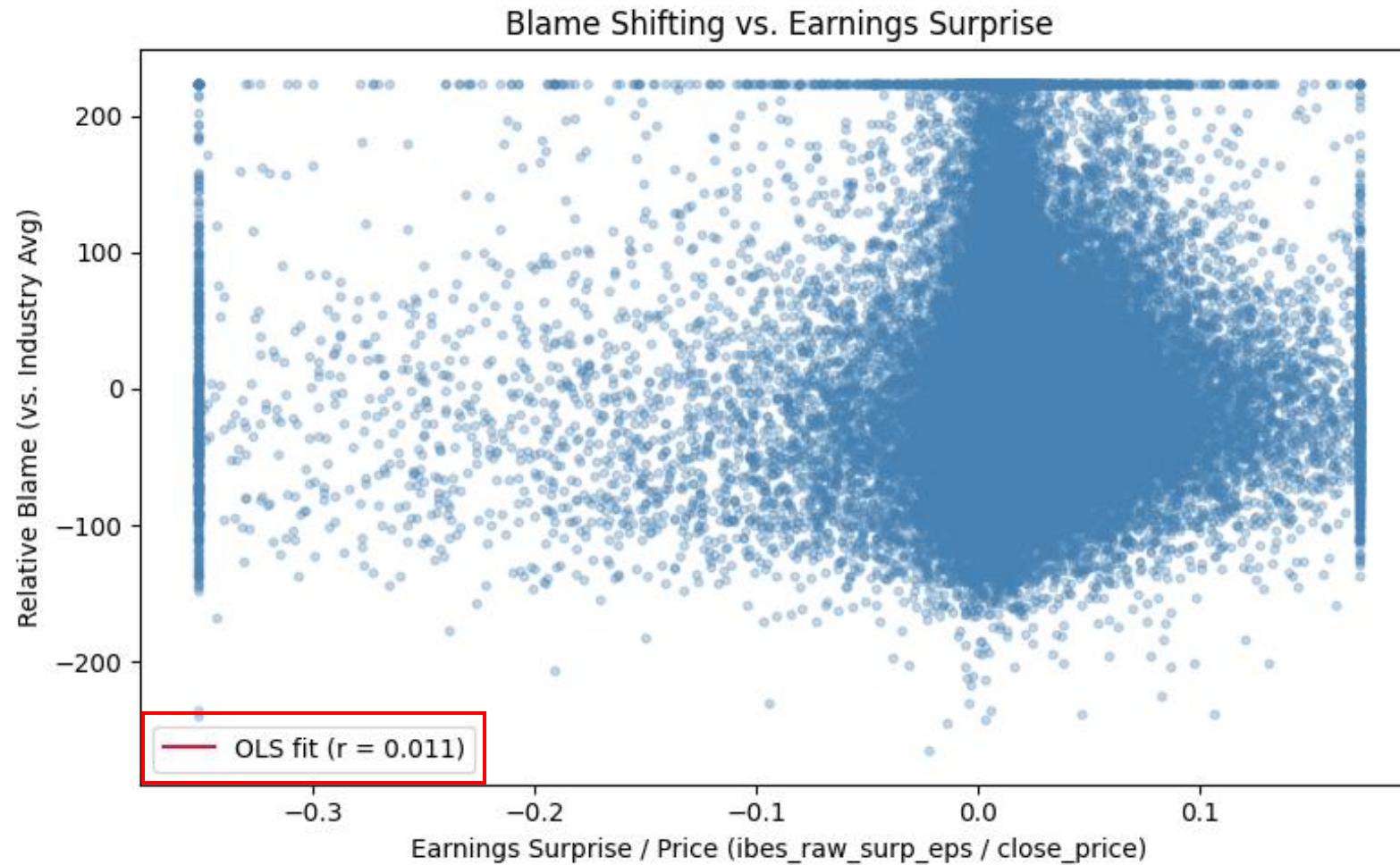
Two internal items drove the shortfall. First, we made the call in August to pause shipments of our Series 7 product for two weeks while we worked through a quality issue identified in our own testing. **That decision pushed roughly \$25 million out of the quarter, and we stand behind it.** Second, sales productivity in our newly hired enterprise cohort is ramping on the timeline we modeled internally, but that timeline contributes less to Q3 than to Q4 and beyond.

Probs : {'negative': 0.9349, 'positive': 0.0651}

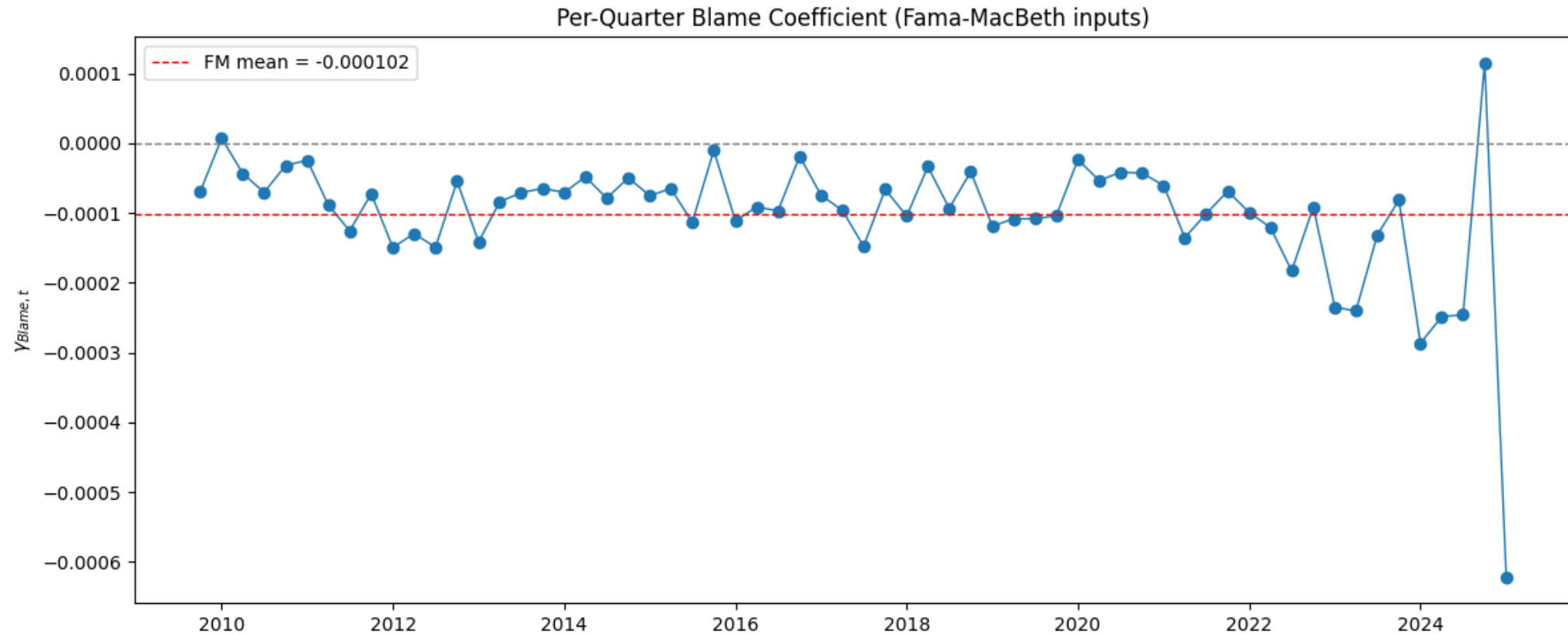
Appendix – Previous Methodology Comparison

EVENT STUDY REGRESSIONS - Dependent variable: CAR(0,+2)				
	(1)	(2)	(3)	(4)
	ES/P + MINUS	ES/P + DIVIDE	ES + MINUS	ES + DIVIDE
Intercept	-0.0021*** (-8.08)	-0.0023*** (-9.06)	-0.0026*** (-9.18)	-0.0028*** (-10.15)
Earnings Surprise	0.0006 (+0.04)	0.0007 (+0.05)	0.0006*** (+6.64)	0.0007*** (+7.04)
Blame Score	-0.0000*** (-8.92)	-0.0000 (-0.28)	-0.0000*** (-8.91)	-0.0000 (-0.28)
N	140600	145242	140600	145242
R ²	0.0026	0.0000	0.0029	0.0003
Adj. R ²	0.0026	-0.0000	0.0029	0.0003
F-stat	39.826	0.039	62.049	24.792
Prob(F)	0.0000	0.9613	0.0000	0.0000
Notes:				
• ES/P = Earnings Surprise / Price[t-1] ES = raw ibes_raw_surp_eps • MINUS = Δfirm_blame - Δavg_FF12_blame DIVIDE = Δfirm_blame / Δavg_FF12_blame • t-statistics in parentheses, HC3 heteroskedasticity-robust standard errors • CAR winsorized at 1% / 99% • Significance: *** p<0.01 ** p<0.05 * p<0.1				

Appendix - Covariance



Appendix – Fama Macbeth





Thank You For An Amazing Semester!

